CSC429: Lab #1

Helen Huy, McCay Rudick, Jess Alencaster

Professor Fang

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1. Open-Source Data Resources

Resource 1: MNIST

Link: https://www.kaggle.com/datasets/hojjatk/mnist-dataset
Description: The MNIST dataset contains 70,000 handwriting digit samples. Frequently used to evaluate Federated Learning models for image classification.

Resource 2: CIFAR-10

Link: https://www.cs.toronto.edu/ kriz/cifar.html
Description: The CIFAR-10 dataset contains 60,000 image samples with 10 classes. Often used to evaluate Convolutional Neural Networks.

2. Open-Source ML Algorithms

Classification: Convolutional Neural Network (CNN)

Link: Keras Sequential Model
Description: A Convolutional Neural Network (CNN) is a deep learning algorithm that takes input features and passes them through convolutional and pooling operations to detect patterns. Very commonly used for image classification.

Regression: Linear Regression

Link: Scikit-Learn Linear Regression
Description: Linear regression is a foundational ML algorithm used to model outcomes based on one or more independent variables using basically a line. Linear regression algorithms are extremely inexpensive and often able to be trained locally for Federated Learning applications.

Clustering: K Means Clustering

Link: Scikit-Learn K-Means
Description: BriK-Means is an unsupervised ML algorithm that partitions unlabelled data into clusters based on feature similarity. Scikit-Learn has an open-source library that is very handy for clustering tasks.

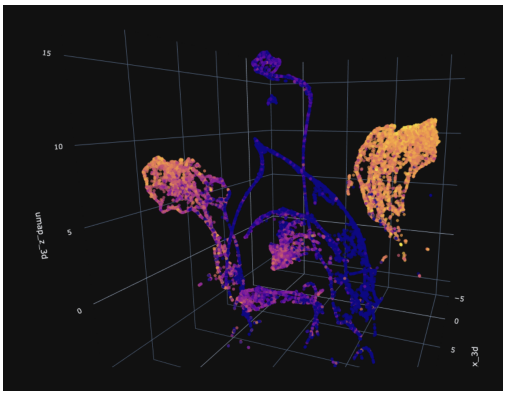

Figure 1: K Means Clustering Example 3d representation

3. Security and Privacy Applications

- **Application 1:** Describe the application briefly and explain why ML is useful for it.
- **Application 2:** Describe the application briefly and explain why ML is useful for it.
- **Application 3:** Describe the application briefly and explain why ML is useful for it.

References

1. Author Last Name, First Initial. (Year). *Title of the article or resource.* Website Name. https://example.com